



From data to decision: Alleviating poverty and promoting development through measuring the unmeasurable economic numbers

Emmanuel A. Onsay^{a,b,*}, Jomar F. Rabajante^a

^a Graduate School, University of the Philippines Los Baños, Laguna 4031, Philippines

^b Partido Institute of Economics, Partido State University, Camarines Sur 4422, Philippines

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ABSTRACT

This work has been carried out and is currently being conducted in the poorest region of Luzon, Philippines. Since poverty is multifaceted and considered unmeasurable in social science, it is notoriously difficult to measure. The methods currently used to measure poverty require a significant amount of time, money, and labor. This poses challenges for policymakers in implementing poverty-reduction policies. Indigenous communities are among the most disadvantaged, vulnerable, and neglected populations in society, facing complex and diverse socioeconomic situations. Poverty stands as one of the oldest and most challenging social problems to have ever existed. Research on indigenous peoples typically takes a qualitative approach, whereas studies on poverty tend to be broad, making them susceptible to significant sampling errors and primarily intended for national policy-making. Using community-based monitoring system (CBMS) data, we achieved a prediction accuracy of 92.60–98.00 % using Random Forest classification and reduced traditional survey and data processing costs by up to 70 %. The proposed model incorporates 27 socioeconomic variables and enables localized policy targeting. Therefore, it is crucial to assess multifaceted poverty and simulate socioeconomic conditions for each tribe to foster economic development. By training and testing datasets, this work proposes new metrics and illustrates the effectiveness of machine learning in predicting poverty. Lastly, the results provide various localities with customized policy targeting tools for poverty alleviation. These techniques can be replicated, adapted, or repurposed by other researchers to assist impoverished populations in improving their well-being.

Social impact

Our research focuses on measuring previously unquantifiable societal issues [1,2]. Understanding the societal implications of this research within economic contexts and among indigenous populations is crucial [3].

This study integrates econometric modeling with machine learning techniques to enhance the accuracy and applicability of poverty forecasts. Traditional poverty research in the country has primarily relied on classic regression analysis, which often limits predictive precision and policy responsiveness. By contrast, our approach generates forecasts that can support more targeted and data-driven interventions for rural communities and marginalized sectors at both disaggregated and national levels. These results offer a framework that can be adapted not only across municipalities in the Philippines but also in similarly situated regions and developing countries worldwide.

Current poverty metrics are often outdated and dependent on costly surveys and censuses that require substantial resources in terms of time and funding. Therefore, the results of this investigation provide an alternative to the conventional reliance on regression-based econometric approaches that have dominated poverty research in the Philippines. By incorporating machine learning techniques, the study introduces predictive and classification models that can uncover complex, non-linear patterns in poverty data, offering more actionable insights for policymakers and development practitioners.

To enhance the accuracy and efficiency of poverty predictions, we have integrated econometrics and machine learning in this work. The resulting predictions can be employed to target policies and programs for rural sectors and localities at both aggregated and disaggregated levels. Furthermore, the government's current analyses are primarily descriptive in nature. By providing diagnostic, predictive, and prescriptive methodologies and models, the output could enhance the data

* Corresponding author at: Graduate School, University of the Philippines Los Baños, Laguna 4031, Philippines.

E-mail address: eaonsay@up.edu.ph (E.A. Onsay).

@EOnsay9489 (E.A. Onsay)

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analytics process in the rural sector.

By utilizing datasets from the community-based monitoring system, this study aids in achieving the objectives of local government units and national statistics agencies by estimating and predicting multidimensional-driven poverty. This implies that the public will have increased opportunities to reduce costs and eliminate non-value-adding operations associated with the annual collection of data and the formulation of policy recommendations for poverty reduction. Moreover, it ensures the efficient collection of data, safeguarding researchers and enumerators in remote, hazardous regions with prevalent social, political, and geographic constraints.

We can share our machine learning algorithms with policymakers, relevant agencies, researchers, and other groups to assist in developing policy-targeting systems aimed at combating poverty at the local levels in the Philippines and globally. The national poverty data are typically holistic and intended for macroeconomic purposes, which may not effectively inform the design of poverty alleviation policies and economic development strategies tailored to specific locales with diverse characteristics. Therefore, we have developed and applied these algorithms specifically for communities and sectors.

In the Philippines, the Bicol region stands out as the poorest region in Luzon, with Camarines Sur identified as the poorest province within Bicol, exhibiting a poverty incidence of 38.7% [1,2]. Through the application of economic analyses, econometric modeling, data analytics, and informatics, our research project aims to capture the multidimensionality of poverty and economic conditions from a microeconomic perspective in Goa, Camarines Sur. Given the multidimensional nature of poverty, it is crucial to measure poverty and inequality across cross-sectional, repeated cross-sectional, and panel data to effectively target programs for poverty reduction and socio-economic development activities for the upliftment of Bicolanos [4].

Data Analytics enables the extraction of technical, practical, computational, and real insights from vast datasets through the science and art of analysis. Data informatics enhances the reliability and validity of analyses through the integration, application, and management of technology, people, and procedures, aiming to measure the unmeasurable in the natural and social sciences [5,6].

This research aligns strongly with the United Nations Sustainable Development Goals, particularly Goal 1 (No Poverty) and Goal 3 (Good Health and Well-being). By employing machine learning and econometric models to measure multidimensional poverty and predict deprivation at the household and community level, we contribute directly to SDG 1 by providing evidence-based tools to accelerate poverty eradication. The proposed policy interventions, ranging from income-generating programs to food security and educational support, are targeted toward the poorest and most marginalized sectors, including indigenous peoples. Moreover, by addressing maternal mortality, child nutrition, and access to basic sanitation and healthcare services, the study also advances SDG 3.

Our localized approach ensures that no one is left behind and provides a scalable model for achieving these goals in similarly impoverished regions worldwide. Recent global studies also underscore the multidimensional implications of poverty alleviation. For instance, Wang et al. explore the decoupling relationship between CO₂ emissions and poverty alleviation in China, suggesting that it is possible to reduce poverty without increasing environmental burdens [11]. Similarly, Zheng et al. examine the global carbon impact of poverty alleviation strategies, highlighting the need for sustainable models that balance development and environmental responsibility [12]. These insights complement our localized, data-driven approach by emphasizing that effective poverty targeting must also consider long-term economic and ecological sustainability, particularly in vulnerable regions like the Bicol area.

Our models are built on publicly available community-based monitoring system (CBMS) datasets and open-source analytical tools, making them highly reproducible and adaptable for use in other municipalities,

provinces, or countries with similar data structures. The methodological framework, including the integration of econometric modeling with machine learning algorithms such as Random Forest, CatBoost, and XGBoost, is modular and can be customized to fit varying local contexts. For instance, researchers and policymakers in other regions can apply our models using their own socioeconomic datasets to identify local drivers of poverty, simulate future scenarios, and target the most vulnerable populations. This reproducibility ensures that the societal benefits of our work are not confined to the Bicol region but can extend to other low-income, disaster-prone, and marginalized communities across the Philippines and globally.

The initial study components and the first phase of the second study have been completed, with policy recommendations now accessible to local government agencies and decision-makers. Table 1 illustrates the research project's Theory of Change (ToC), hypotheses, findings chain, and societal impact in the poorest part of Luzon, Philippines. The study's Theory of Change aligns with frameworks such as Health and Nutrition ToC, Education ToC, Social Justice ToC, Growth ToC, and State-Society Relation ToC [7].

Fig. 1 illustrates vital results for societal impact assessment. Measuring poverty in a specific locality with various structures is crucial for economic development. The sample results, as shown in Fig. 1, demonstrate that the proportion of individuals and households living below the poverty line and food threshold varies by locality. Poverty outcomes are significantly influenced by multidimensional variables. Moreover, poverty in each locality is affected by intervening variables. Each examined model is relevant in all areas, accurately predicting the outcomes. While the 12 indigenous tribes exhibit varying levels of poverty, most households are below the poverty line. The probability of typhoons, floods, droughts, landslides, or mudslides significantly affects an individual's readiness for disaster risk. The frequency of natural disasters is correlated with preparedness. Typhoons are the most prevalent and noticeable hazard in the region. Therefore, precautionary measures should be implemented before, during, and after such calamities.

We integrated advanced econometric models with machine learning-based regression and classification techniques to analyze poverty indicators across multiple sectors and communities. As illustrated in Fig. 2, the study conducted 273 ensemble runs for regression and 468 for classification, utilizing seven regression and twelve classification algorithms. These were applied to datasets representing 34 localities and four development sectors, with results cross-validated prior to integration. Among the models tested, the random forest classifier yielded the highest classification accuracy, while the random forest regressor delivered the most robust predictive performance. These results demonstrate consistent and statistically significant relationships between poverty outcomes, measured through incidence, gap, severity, and Watt's indices—and a set of 27 multidimensional variables. Importantly, the findings highlight distinct configurations of poverty across different localities, allowing for the development of finely tuned, context-specific policy tools. This enables more precise targeting of poverty alleviation programs and supports the design of data-informed interventions for economic development, particularly in marginalized and indigenous communities.

Table 2 presents the recommended policy interventions derived from this research, aimed at addressing key drivers of poverty and supporting localized economic development. Policymakers and relevant agencies are encouraged to consider these evidence-based findings for targeted implementation. The modeling approach employed in this study is designed to be replicable, offering a framework that other researchers and institutions can apply in different contexts to generate comparable insights. Residents were grouped into three rounds based on the prevalence of multidimensional poverty, using a-priori variables to identify the most vulnerable populations. Priority for intervention should be given to individuals in Round 1, who exhibit the highest levels of deprivation. This tiered strategy allows for more efficient allocation of resources by focusing on the most disadvantaged groups, ultimately

Table 1

Theory of change (ToC), hypotheses, and results chain of the research project societal impact in the poorest region of Luzon, Philippines.

Objectives	• Measure poverty and evaluate inequality in all barangays and sectors of Goa, Camarines Sur. • Analyze the relationship between health and nutrition, housing and settlement, water and sanitation, education, livelihood and employment, peace and order, calamity occurrences, disaster risk preparedness, and poverty and inequality outcomes. • Predict poverty and socioeconomic conditions through machine learning algorithms. • Measure Chronic and Transient Poverty in Goa, Camarines Sur, the Philippines. • Describe the Characteristics of Chronic and Transient Poor Households. • Examine the impact of calamity occurrences, disaster risk preparedness, and economic shocks on Chronic and Transient poverty. • Examine the determinants of Chronic and Transient poverty. • Measure poverty and evaluate inequality among the indigenous people of Mt. Isarog. • Propose poverty alleviation programs, policy efforts, and intervention initiatives for the economic and community development of Goa municipality. • Propose poverty alleviation programs, policy efforts, and intervention initiatives for the economic and community development of indigenous people in Mt. Isarog																														
Input	❖ Budget and financial resources in the form of research grants; ❖ Human resources such as researchers, policy implementers, enumerators, editors, and staff; ❖ Physical resources such as facilities, supplies, and materials, and ❖ Time resources from planning to implementation.																														
Processes	■ Coordinating with various partners or linkages by the academe; ■ Establishing the institution or program; ■ Data curation, cleaning, wrangling, and coding; ■ Model fitting, diagnostic tests, and validity tests; ■ Econometric modeling and machine learning algorithm runs; ■ Data analytics and prescriptive policy proposal making; ■ Dissemination, presentation, and publication of research; ■ Adoption and implementation of policy proposals; ■ Audit of financial resources, and monitoring and evaluation of the programs.																														
Outputs	• Improved human capital conditions by enhancing health (regular check-ups, adequate health prescriptions, dropped illness records, increased immunization, normal BMI, etc.) and education status (regular class attendance, enhanced academic performance, short-run labor activity, etc.); • Increased visitation of pregnant mothers to health clinics in regular basis. • Increased participation of children in school. • Enhanced household income and management of risk (With household challenges like consumption/expenditure, households become more flexible and adaptable); • Increased investment in human capital and physical assets leads to resilience (beneficiaries become more aware and confident about their own and their families' microeconomic problems).																														
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enhancing the effectiveness and cost-efficiency of poverty alleviation programs. A broader policy application targeting all municipalities and vulnerable populations is summarized in [Table 3](#).

We developed a policy targeting framework that categorizes localities into high, moderate, and low priority groups across key development indicators such as child nutrition, maternal mortality, informal housing, sanitation, school enrollment, unemployment, and food insecurity. This categorization enables more efficient program delivery and resource allocation by focusing interventions where needs are most critical. Using data from 34 localities, the model applies a three-round classification system based on prevalence intensity, allowing local governments to adjust strategies dynamically as new data becomes available.

While our case study focuses on municipalities in the Philippines, the framework is designed for replication. It provides a practical template for policymakers, development agencies, and researchers to implement evidence-based poverty targeting in other developing or resource-constrained settings [8]. By introducing modular variables and applying longitudinal tracking, the system supports dynamic monitoring of household- and individual-level poverty transitions, which are essential for the design, evaluation, and refinement of social protection

and economic development programs over time.

Methodology

We propose new methods to measure the unmeasurable multifaceted poverty deprivations and socioeconomic conditions. This includes: a comprehensive statistical analysis to examine socioeconomic situations using diagnostic and descriptive analytics; a blend of advanced econometrics and predictive analytics to gauge multidimensional poverty; and the incorporation of machine learning to model socioeconomic scenarios and prescriptive analytics to formulate policies [1,2].

We have applied the following methods:

1. Statistical and Regression Modeling:

- a. Descriptive Analytics (involves summarizing historical poverty data to describe past occurrences and understand the current state of socioeconomic conditions.)
- b. Diagnostic Analytics (focuses on determining why certain events happened by delving into data to identify patterns and root causes.)

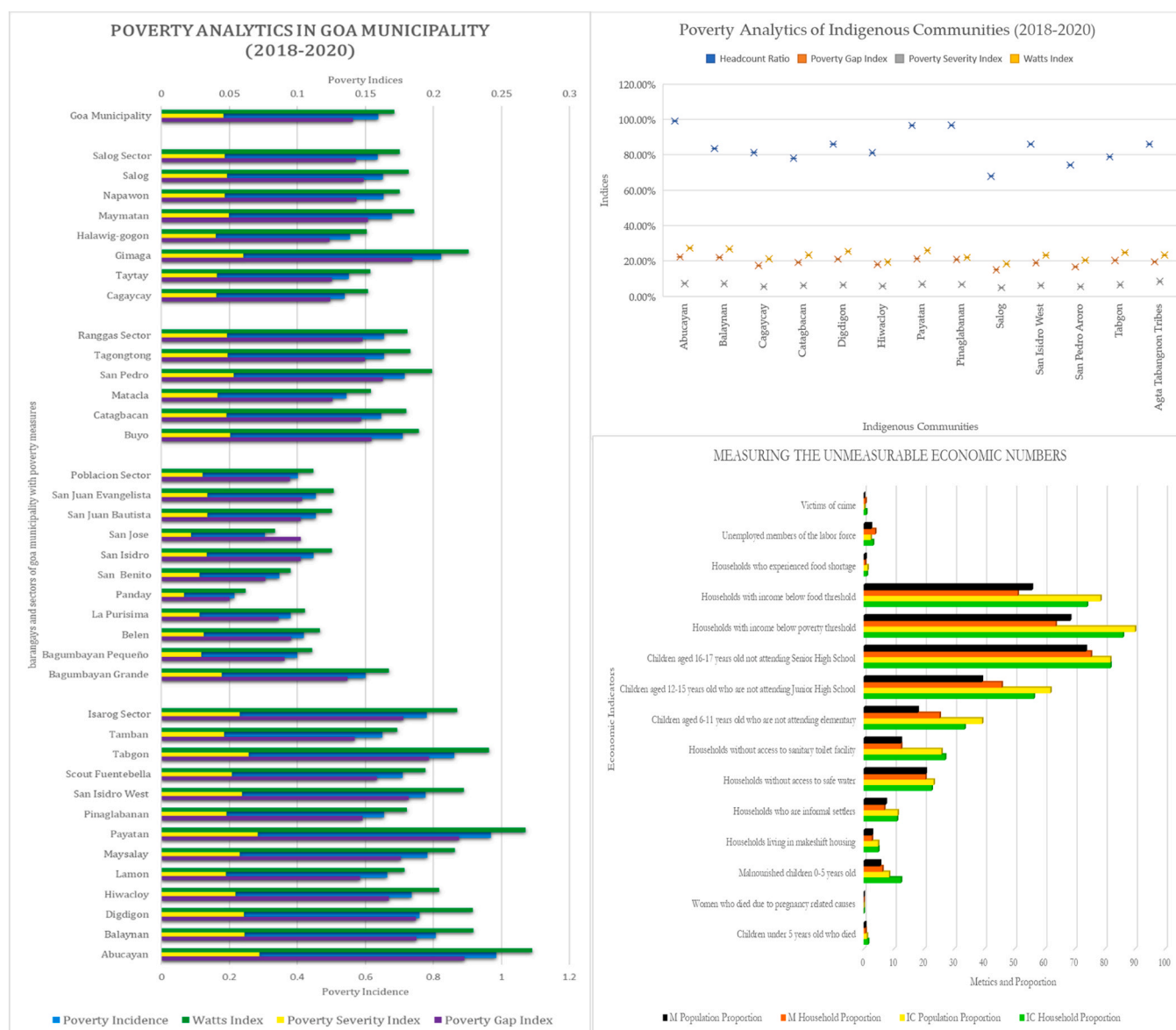


Fig. 1. Sample results of statistical and econometric analyses in measuring the unmeasurable economic numbers.

- c. Multidimensional Analysis (involves examining data across multiple dimensions or attributes to gain a comprehensive understanding of relationships and trends.)
- d. Logit Regression (was used to model binary outcomes or events based on predictor variables.)
- e. Probit Regression (similar to logit regression that models binary outcomes using a different set of assumptions under normal distribution.)
- f. Statistical Classification (involves categorizing data points into pre-defined classes or categories based on their attributes or features.)
- g. Predictive Marginal Analysis (focuses on estimating the impact of individual variables on outcomes while holding other variables constant.)
- h. Goodness-of-fit tests (assess how well a statistical model fits the observed data, indicating the model's adequacy in representing the data.)
- i. Data Visualization (refers to the graphical representation of data to aid in understanding patterns, trends, and insights within the data.) [1,2].

Statistical and regression modeling provides a comprehensive

framework for poverty analysis by combining descriptive, diagnostic, and multidimensional analytics with logit and probit regressions, classification, and predictive marginal analysis. These methods uncover patterns, model binary outcomes, and estimate variable impacts, while goodness-of-fit tests and data visualization ensure model accuracy and clarity of insights.

2. Poverty Analytics and Econometrics:

- a. Poverty Incidence Analysis (involves examining the proportion of individuals or households living below the poverty line within a given population.)
- b. Poverty Gap Metrics (quantify the average income shortfall of the poor from the poverty line, providing insight into the depth of poverty within a population.)
- c. Poverty Severity Statistics (measure the intensity of poverty experienced by individuals or households below the poverty line, offering a more nuanced understanding of the extent of deprivation.)
- d. Watts Index Decomposition (dissects the changes in poverty levels over time or across different groups, attributing these changes to various factors or components.)

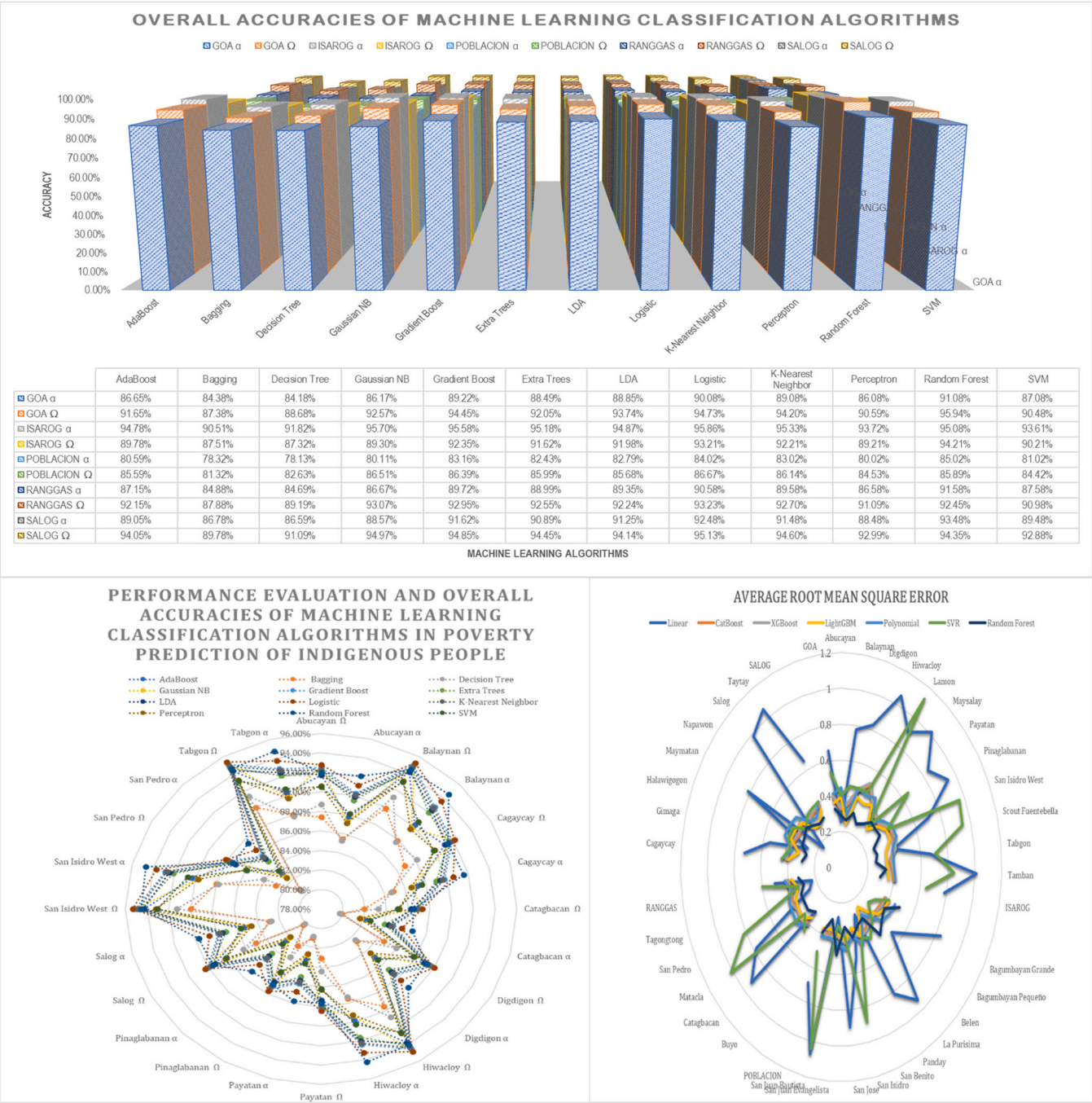


Fig. 2. Results of machine learning regression and classification for poverty prediction.

e. Data Visualization (Data visualization refers to the graphical representation of poverty-related data to enhance understanding and communicate insights effectively.) [1,4].

Poverty measurement involves analyzing the incidence, depth, and severity of poverty using key metrics. Poverty incidence reveals the share of the population below the poverty line, while poverty gap and severity statistics capture the depth and intensity of deprivation. The Watts Index decomposition further disaggregates changes in poverty across time or groups. Data visualization enhances interpretation, making complex poverty patterns more accessible for policy and planning.

3. Machine Learning Regression:

- a. CatBoost (gradient boosting algorithm that is specifically designed to handle categorical features efficiently.)
- b. XGBoost (eXtreme Gradient Boosting is a powerful algorithm that excels in speed and performance.)
- c. LightGBM (Light Gradient Boosting Machine is a framework that uses a tree-based learning algorithm.)
- d. Polynomial Regression (a form of regression analysis where the relationship between the independent variable and the dependent variable is modeled as an nth degree polynomial.)
- e. SVR (Support Vector Regression is based on the Support Vector Machine technique, used for predicting continuous variables.)
- f. Random Forest (an ensemble learning method that constructs a multitude of decision trees during training and outputs the average prediction of the individual trees for regression tasks.) [9].

Table 2

Proposed intervention programs and policy initiatives for economic development of the region.

Poverty Indicators		Intervention Programs and Policy Initiatives
Nutrition and Well-being	Child Fatality	- Monitoring and Supervising Sick or Ill Children - Conducting Training Sessions on Healthy Nutrition and Food Handling - Information, Education, and Communication (IEC) Initiatives and Resources Promoting Proper Hygiene, Health, and Sanitation - Regular Home Visits by Healthcare Providers for Health Checks and Supplements for Children Aged 0–5 - Nutritional Support Program to Address Malnutrition - Implementing Household-Level Food Cultivation - Enforcing Laws, Regulations, and Monitoring Impact to Minimize Risks and Decrease Mortality Rates in the Community
	Maternal Mortality	- Rigorous monitoring of expectant mothers for their routine appointments and immunizations - Enforcing the Municipal Ordinance mandating Facility-Based Deliveries (banning home births)
	Nutritional Deficiencies in Children	- Setting targets for achieving all four Antenatal Care (ANC) sessions - Nutrition Assistance Program - Parenting Workshops - Health Assessments and Dietary Supplements - Health Advisory Sessions - Food Cultivation Initiative (backyard organic gardening supported by local regulations and impact assessments)
Residences and Land Settlement	Temporary Dwellings	Selection of a suitable relocation area for residential projects with livelihood opportunities and essential service access.
	Informal Housing Arrangements	- Supply of housing units for residents in informal settlements. - Implementation of a rent-to-own housing initiative in collaboration with HUDCC and relevant agencies for qualified recipients.
Hygiene and Water	Households without access to clean drinking water	- Adoption and rigorous enforcement of Barangay Zoning Regulations. - Facilitate Watershed Development Workshops - Information, Education, and Communication (IEC) Campaigns on Health and Sanitation - Testing and Treating Solar Water Sources - Installation of Safe Water Facilities
	Households without access to a hygienic sanitary restroom	- Supply of Hygienic Toilets - Enforcement of Health and Sanitation Ordinances
Fundamental Education	Kids who are not enrolled in elementary school between the ages 6–11	- Feeding programs for Children - Provision of School supplies and uniforms to identified children age 6–11 not attending elementary - Implementation of Ordinances for the members of the 4Ps - Information Education and Communication (IEC) on literacy campaign - Ordinances/local legislation/policies - Various support to teachers and learners - Partnership with academic and private institutions for education campaigns and extension programs
	Children who are not enrolled in junior high school between the ages of 12 and 15	- ALS Program - TESDA Skills training - Policies/Ordinances - Various support to teachers and learners - Partnership with academic and private institutions for education campaigns and extension programs
	Children who are do not attend senior high school between the ages of 16 and 17	- ALS Program, - TESDA Skills training - Policies/Ordinances - Various support to teachers and learners - Partnership with academic and private institutions for education campaigns and extension programs
Earnings and Subsistence	Poverty based on Income Threshold	Backyard gardening
	Poverty based on Food Threshold Food Shortage Unemployment	- Proper technology on farming and impact monitoring of programs and projects (DA) - Ordinance on backyard gardening - Women empowerment (Livelihood programs) - Technical Skills training (TESDA) - Advocacy training on Food preparation/processing - Ordinances/local legislation/policies - Partnership with academic and private institutions for skills training and extension programs
Harmony and Stability	Victims of crime	- CCTV (high definition with registered plate recognition) - Profiling of all the migrants to our municipality within five years - Strict implementation of Curfew hours - Local Ordinance on Excessive use of gadgets and social media

Table 3

Multidimensional poverty indicators, policy interventions, and target areas for localized development.

Indicator Domain	Key Indicators	Recommended Policy Interventions	Target Groups/Priority Areas
Health and Nutrition	Child malnutrition, maternal health risks, food insecurity	Community nutrition programs, maternal health clinics, food subsidy schemes	Underserved rural and indigenous communities
Housing and Sanitation	Informal settlements, inadequate sanitation, unsafe water	Affordable housing projects, WASH (Water, Sanitation, and Hygiene) initiatives, infrastructure support	Urban poor and resettlement communities
Education	High dropout rates, low enrollment, limited school access	Conditional cash transfers for schooling, mobile learning units, school feeding programs	Marginalized youth, remote and rural learners
Employment and Livelihood	High unemployment, informal labor, income instability	Livelihood development, vocational training, access to microcredit	Informal workers, displaced and low-income households
Disaster Risk and Resilience	Exposure to climate hazards, flood risk, weak preparedness	Community-led DRR (Disaster Risk Reduction), early warning systems, climate-resilient planning	Coastal zones, forest zones, and high-risk settlements
Crime and Social Vulnerability	Crime prevalence, gender-based violence, social insecurity	Community-based policing, gender protection initiatives, psychosocial support services	Conflict-affected, high-risk, and vulnerable communities

Advanced machine learning methods enhance poverty prediction by leveraging both structured and unstructured data. Algorithms like CatBoost, XGBoost, and LightGBM offer high-performance gradient boosting approaches, particularly effective with categorical and large-scale data. Polynomial regression and SVR model complex, nonlinear relationships for continuous outcomes, while Random Forest uses ensemble decision trees to improve predictive accuracy. These techniques enable more precise, data-driven insights into poverty dynamics.

4. Machine Learning Classification:

- AdaBoost (an ensemble learning method that combines multiple weak classifiers to create a strong classifier.)
- Bagging (an ensemble method that builds multiple models independently and then aggregates their predictions.)
- Decision Tree (a tree-like model of decisions and their possible consequences)
- Gaussian Naïve Bayes (a probabilistic classifier based on applying Bayes' theorem with strong (naive) independence assumptions between the features.)
- Gradient Boost (an ensemble technique that builds models sequentially, with each new model correcting errors made by the previous ones.)
- Extra Trees (an ensemble learning method similar to Random Forest, but with additional randomness in the feature selection process.)
- LDA (Linear Discriminant Analysis is a method used for dimensionality reduction and classification tasks.)
- Logistic Regression (a statistical model used for binary classification that estimates the probability of a certain class.)
- K-Nearest Neighbor (stores all available cases and classifies new cases based on a similarity measure.)
- Perceptron (an artificial neural network used for binary classification tasks.)
- Random Forest (an ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of the individual trees for classification tasks.)
- Support Vector Machine (a powerful supervised learning algorithm used for classification tasks.) [9].

This set of classification algorithms and ensemble methods supports robust predictive modeling in poverty analysis. Ensemble techniques like AdaBoost, Bagging, Gradient Boost, Extra Trees, and Random Forest improve accuracy by combining multiple models. Core classifiers such as Decision Tree, Logistic Regression, K-Nearest Neighbor, Gaussian Naïve Bayes, and Support Vector Machine offer diverse approaches to pattern recognition. Advanced methods like LDA and Perceptron enhance classification and dimensionality reduction. Together, these tools enable precise and scalable identification of poverty-related patterns across complex datasets.

5. Prescriptive Analytics:

- Policy Mapping (involves identifying and visualizing the relationships between policies and their potential impacts on various

outcomes.)

- Targeting Analysis (focuses on identifying specific target groups or areas where interventions or policies should be directed to achieve desired outcomes.) [2].

Prescriptive analytics guides decision-making by identifying optimal actions through policy mapping and targeting analysis. Policy mapping visualizes the links between interventions and expected outcomes, while targeting analysis pinpoints priority groups or areas for focused implementation. Together, they support data-driven, effective, and context-specific poverty alleviation strategies.

6. Inequality Analysis:

- Decile Dispersion Ratio (measures the difference between the average income of the top and bottom deciles of a population.)
- Gini Coefficient of Inequality and Lorenz Curve (The Gini coefficient quantifies the level of income inequality in a population, while the Lorenz curve visually represents income distribution.)
- Generalized Entropy Measures (measures income distribution by considering the entire distribution rather than just the mean.)
- Atkinson's Inequality Index (measures economic inequality that incorporates society's aversion to inequality.)
- Pen's Parade (a graphical representation used to illustrate income distribution and inequality, particularly in relation to the Lorenz curve.) [4].

Income inequality analysis uses various metrics to capture disparities across populations. The Decile Dispersion Ratio highlights income gaps between extremes, while the Gini Coefficient and Lorenz Curve quantify and visualize inequality. Generalized Entropy Measures and Atkinson's Index offer deeper insights by considering the full distribution and societal preferences. Pen's Parade provides an intuitive graphical view of income stratification. Together, these tools offer a comprehensive understanding of inequality dynamics.

7. Santigwar Poverty Measurement:

- Localized Poverty Measurement (involves assessing poverty levels at a specific geographical scale)
- Composite Indexing (combines multiple indicators or variables into a single index to provide a comprehensive measure of poverty in a local scale.)
- Clustering Systems (group similar areas or individuals together based on poverty indicators, allowing for targeted interventions and analysis in specific locale.) [1,2].

The Santigwar Poverty Measurement framework emphasizes localized, data-driven approaches to assessing poverty. It utilizes localized poverty measurement to capture conditions at the community level, composite indexing to integrate multiple indicators into a single poverty score, and clustering systems to group similar areas or households for more targeted and effective interventions. This approach supports

precision policymaking tailored to specific local contexts.

8. Longitudinal Analysis:

- a. Repeated Cross-sectional Regression Modeling (involves analyzing data from multiple cross-sectional surveys conducted at different time points to study relationships between variables over time.)
- b. Panel Logistic Econometrics (applies logistic regression models to panel data, which involves observations on multiple subjects over multiple time periods, to analyze relationships and make predictions.)
- c. Poverty Exits (statistical computation of the number of years required for individuals or households to transition out of poverty). [1,8,10].

Longitudinal analysis tracks poverty dynamics over time using repeated cross-sectional regression and panel logistic econometrics to examine changing relationships and predict trends. It also includes measuring poverty exits, estimating the duration needed for individuals or households to move out of poverty. This approach provides deeper insights into the temporal aspects of poverty and the long-term impact of interventions.

9. Impact Evaluation:

- a. Randomized Control Trials (participants are randomly assigned to either a treatment group or a control group to measure the impact of an intervention or treatment).
- b. Instrumental Variable Regression (address endogeneity issues by finding instrumental variables that are correlated with the treatment but uncorrelated with the outcome).
- c. Regression Discontinuity Design (a quasi-experimental design where participants are assigned to treatment or control groups based on a cutoff point in a continuous variable to estimate causal effects [7]).

Causal inference methods such as Randomized Control Trials, Instrumental Variable Regression, and Regression Discontinuity Design enable robust estimation of intervention impacts. RCTs ensure unbiased comparisons through random assignment, IV regression addresses endogeneity by using external instruments, and RDD leverages cutoff-based assignments to estimate causal effects. Together, these methods strengthen the validity of poverty-related policy evaluations.

Machine learning algorithms are employed in this study to analyze diverse data points, such as income, education, health indicators, and demographics to predict the likelihood of individuals or communities experiencing poverty. Using community-based datasets, the models are trained on historical data to identify complex, non-linear relationships that traditional methods may overlook [1,2]. Once trained, these models can generate predictions for new, unseen populations, enabling policymakers to target interventions more precisely, allocate resources more efficiently, and tailor poverty alleviation programs to specific community needs.

While this work uses data from the Philippines' community-based monitoring system, the methodology is adaptable to other regions with similar local census data. The study introduces a sequence of methods and techniques that can be replicated or modified by researchers, planners, and development agencies globally—particularly in low-income and developing countries. Through rigorous data processing, including cleaning, filtering, and feature engineering, the resulting insights offer benchmarks for economic analysis and poverty targeting.

To evaluate the impact of data-driven policies, the study also proposes monitoring frameworks using treatment and control groups, enabling assessments of causal effects and changes over time [8,10]. This shift from raw data to informed decision-making exemplifies how machine learning can drive transformative change in public policy, social planning, and poverty reduction efforts worldwide.

Results and implications

This study demonstrates how machine learning-generated poverty maps can support more efficient targeting of interventions at the local level. By identifying areas with high poverty prevalence and populations vulnerable to transient poverty, development efforts such as maternal health outreach, school feeding programs, and disaster preparedness can be more data-driven and responsive. This approach reduces misallocated aid and ensures that the most at-risk households receive timely support. The methodology can be adapted by governments and organizations worldwide to enhance the efficiency of localized policy implementation.

Our findings, supported by statistical, econometric, and machine learning analyses, offer new ways to capture and predict poverty patterns. Visual summaries in this study illustrate how data can inform policy design, guide the implementation of economic interventions, and validate key relationships between poverty, inequality, and marginalized populations. While developed within a Philippine context, these tools are applicable to other developing regions seeking scalable, evidence-based solutions for economic development and poverty reduction.

Machine learning regression and classification models presented in this work provide a replicable framework for enhancing poverty alleviation strategies. These tools support data-informed decision-making and can improve the precision of program targeting. Once completed, the full dataset and methodological documentation will be made available to assist governments, researchers, and development agencies in replicating or adapting the approach for broader use.

CRedit authorship contribution statement

Emmanuel Onsay: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Visualization, Project administration, Funding acquisition. **Jomar F. Rabajante:** Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

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Ethical statements

The conduct of this study was approved by Partido State University and the University of the Philippines Los Baños. The Goa, Camarines Sur Local Government Unit provides authorization for the results of the processed data to be used, manipulated, analysed, and distributed. Researchers voluntarily conduct the data analysis and procedures; they are not a component of any experiment. Our human data are secondary, indirect, and socioeconomic in nature; they don't entail any laboratory work. Thus, ethical clearances are not required. Furthermore, this study does not involve any animal testing, direct human volunteers, or data acquired from social media platforms.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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